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Agent-based simulation of building evacuation: Combining human behavior with predictable spatial accessibility in a fire emergency

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ABSTRACT

The building environment and evacuation behavior are decisive factors for building evacuation performance. Shifting toward the use of agent-based models, many current studies have considered the heterogeneous evacuation behavior resulting from an individualized perception of the building environment, but few studies have incorporated the evacuees' awareness of the predictable change in the spatial accessibility by activated fire safety facilities during emergency scenarios. To investigate the specific influence of such spatial change on the evacuation performance, this study presents an agent-based building evacuation model in which the evacuee's knowledge, including both the spatial knowledge of the stationary environment during a normal situation and the event knowledge of the predictable spatial change for fire-fighting purposes, is considered. In addition, a semantic representation of building environment is developed to represent the alterable connectivity structure when considering the fire safety facilities. Using the proposed model, a series of evacuation simulations have been conducted for groups of evacuees with different knowledge levels during three specific fire scenarios. The simulation results suggest that the proposed model can evaluate the potential influence of the spatial change on the evacuation efficiency, which is dependent on the evacuees' knowledge level and the location of the fire safety facilities. The model, although a prototype at this stage, will facilitate more realistic evacuation simulation in fire emergency scenarios and will support building evacuation management.

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1. Introduction

Efficient emergency evacuation is important for building fire safety. The inner space of a building is divided by physical barriers into enclosures such as rooms and corridors that are connected to each other through openings such as doors and windows. The connectivity structure of these enclosures determines the spatial accessibility throughout the building and is a crucial factor for building emergency evacuation [25]. When a fire emergency occurs in the building, the connectivity structure might be changed for fire-fighting purposes. According to the Hong Kong building fire safety code [4], a large building should be divided into fire compartments. Separated parts, such as fire rolling shutters, will be activated according to the

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location of the burning point to resist the passage of fire and toxic gasses to or from another part of the building. However, the activated separated parts also break the spatial connectivity between the adjacent spaces and, consequently, some escape routes are blocked. Therefore, it is necessary to consider the change in the connectivity structure by activated fire safety facilities during emergency situations, which usually leads to different but predictable spatial accessibility.

The evacuees' knowledge of spatial accessibility may greatly influence the evacuation efficiency [21]. Taking an office building for instance, the staff could be familiar with every place and the connectivity structure inside the building, while the public may only know the main entrance and the particular route through which they came in. As people tend to attach to familiar places during the evacuation [44], the staff and the public may follow different escape routes which are not necessarily the shortest route. Moreover, the staff who have attended fire drills could be aware of the potentially changed spatial accessibility due to activated separated parts and avoid the blocked routes in advance. In contrast, the public who lack awareness of the activated separated parts may walk into a blocked route causing U-turns and 'turbulence'. Therefore, in order to assess the evacuees' capability of escape and provide critical evacuation assistance, it is important to consider not only the evacuees' knowledge of the spatial accessibility during normal situation but also their awareness of the predictable spatial change regarding the fire safety facilities.

Given the above mentioned facts, the purpose of this study was to simulate building evacuation with a focus on the potentially changed spatial accessibility by fire safety facilities. To reach this goal, we developed a semantic model to represent the alterable connectivity structure within a building regarding the fire safety facilities. The evacuee's knowledge, including not only the spatial accessibility during a normal situation but also the predictable change by the activated fire safety facilities during an emergency situation, is considered in the evacuation model. This approach is expected to predict a more reliable building evacuation performance and thus provide a more rational basis for evacuation management. The paper is organized as follows: In the next section, we briefly review the related work in the field of building evacuation simulation. In Section 3, a framework of the proposed evacuation model is presented considering the predictable change in the connectivity structure and the evacuees' knowledge of the predictable spatial accessibility. The detailed implementation method is described in Section 4, and in Section 5, the evacuation simulation of a campus building is performed with a discussion of the simulation results. The conclusions are given in the final section.

2. Related work

The building environment and human behavior are the two critical scenario variables in real-world building evacuation [21]. Focusing on these two variables, the essential task of evacuation simulation is to represent the evacuee's interaction with the building environment [11].

Pioneering work in building evacuation simulation mainly focused on the fixed spatial constraints of the building environment with the purpose of optimizing the evacuees' movement and minimizing the evacuation time. This focus is particularly true for models such as EVACNET4 [8,20], EESCAPE [19] and Exit89 [5], where the connectivity structure of the building is modeled as a fixed network. However, the assumption that the spatial accessibility within the building remains the same could be easily violated in a fire emergency situation as exits or routes might be blocked due to certain conditions (e.g., smoke or flames). Models such as FDS + Evac [23] and STEPS [27] have considered the dynamic availability of exits or routes during emergency situations. Furthermore, in order to depict reliable environmental conditions, fire simulators have been coupled with evacuation models [10,12,46]. However, the approaches adopted to represent the building environment, including the coarse network, the fine network and the continuous space [24] are mainly concerned with the dynamic spatial accessibility, if any, in terms of environmental conditions or manual control by the user. Functional feature of the building, such as the fire safety facilities, and its influence on spatial accessibility has not been fully considered.

The evacuee's behavior can be represented in many different ways. Earlier models simulate the evacuees as a continuous homogeneous mass that behaves as a fluid flowing along the corridors. The speed of the fluid flow is calculated as a function of the density subject to the geometric features of the environment [9,33,36]. In fact, the evacuee's movement could be significantly influenced by interactions with obstacles such as walls and other evacuees. In view of this, models such as the magnetic model [31], social force model [15], floor fluid model [2] and multi-grid model [45] simulate the evacuees as homogeneous particles moving around as an emergent function of attractive force, repulsion force and friction force regarding the interaction with the surrounding environment. In order to reproduce heterogeneous evacuation behaviors, agent-based simulation has been widely used [42,50]. The agents may have different familiarities with the building and choose their own escape route which is not necessarily the shortest path [32,34,48]. The impact of exit's visibility on route selection has been highlighted [7,22], especially in smoke-filled environment [30,41]. Moreover, individual differences in route selection regarding the dynamic spatial accessibility due to congestion, smoke or flames have been considered. Agents may follow a self-estimated quickest route [18] and decide whether to redirect away from the danger [12] depending on their physical and psychological characteristics. However, these studies do not fully describe individualized route choices in relation to the evacuees' awareness of the predictable change of spatial accessibility when certain fire safety facilities are activated.

From a review of previous work, we noticed that there is growing concern for the dynamic spatial accessibility during emergency situations and heterogeneous evacuation behavior resulting from the individualized perception of the building environment [11,24,35,50]. However, the existing models are primarily concerned with the evacuees' knowledge of fixed

spatial features during a normal situation, as well as with the instinctual reactions to the dynamic spatial accessibility affected by environmental conditions (e.g., smoke and flames). Evacuees' awareness of the predictable spatial accessibility that is changed by the activated fire safety facilities during emergency situations and the consequential route adjustment seem to be neglected. Aiming to provide a more reliable prediction of the evacuation performance, it is necessary to develop a building evacuation model that combines the evacuees' behavior with the predictable spatial accessibility during emergency situations.

3. Model framework

This study mainly concentrates on the effect of activated fire safety facilities (e.g., the fire rolling shutter) on the evacuation performance during a building fire emergency. We developed an agent-based modeling framework that combines the evacuee's behavior and the predictable spatial accessibility that is changed by the fire safety facilities (Fig. 1). The basic principle of agent-based modeling is to create emergent phenomenon by simulating individual-level behavior [6,43] and this modeling approach has been widely used in many areas for behavioral analysis [1,3,16]. Agent-based evacuation models usually consist of autonomous agents (evacuees) that perform heterogeneous evacuation behaviors within a virtual space (the building environment).

To simulate the individualized route selection under the influence of a potentially changed connectivity structure, the evacuees' knowledge of the spatial environment, including the knowledge of the stationary spatial features and the awareness of the activated fire safety facilities, are considered. Each agent will select their own escape route based on the assumed spatial accessibility. When congestion occurs or the route is blocked, the agent will update its knowledge and adjust the escape route if there is another alternative route.

Regarding the representation of the building environment, the static geometric features can be extracted from the floor plan. To represent the change in the connectivity structure by the fire safety facilities, a semantics-based topological representation is hereby proposed in the context of building fire safety that allows the evacuation model to dynamically update the connectivity of the internal spaces, such as rooms, hallways and corridors, according to the state of the fire safety facilities. To simulate the evacuees' local movement within each internal space, the building environment is geometrically represented as a grid consisting of small cells. Cells overlapping with stationary obstacles such as walls and closed doors are labeled as occupied cells, which are indicated by the semantic representation and updated when the fire safety facilities are activated. In addition, each agent is a moving obstacle for others, and the occupancy state of underlying cells will be updated at each movement.

4. Model implementation

4.1. Semantic model for alterable accessibility during a fire scenario

4.1.1. Semantic ontology for a building fire emergency

Models incorporating semantic information have been highlighted in constructing building interior spaces. The Building Information Model (BIM) was proposed as the international industry semantic standard, and is capable of storing both the geometric and semantic information of the building environment [29]. Insufficient efforts have been devoted to constructing semantic models of interior spaces for building evacuation [28]. However, semantic ontologies in the context of building fire safety and their hierarchical relationships have not been clearly identified in previous studies [40]. Concentrating on

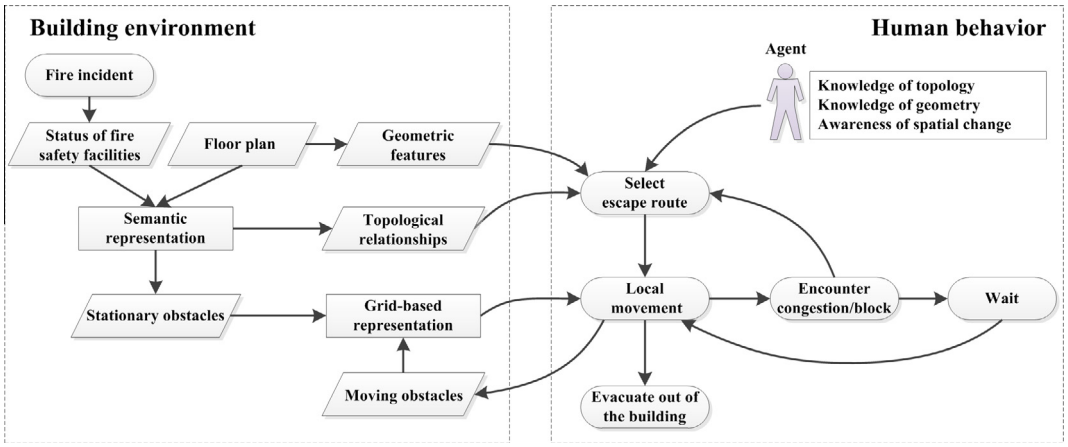


Fig. 1. The framework of agent-based building evacuation simulation combining human behavior with predictable spatial accessibility.

the fire emergency scenario, we expand building semantic ontology on fire safety based on the semantic information described in the BIM, thereby considering the change in the connectivity structure by fire-fighting measures when performing building evacuation simulation. Three major semantic ontologies are identified including fire compartment, building opening and building component (see Fig. 2) and are defined below.

(1) *Fire compartment:* A fire compartment is a space within the building that is enclosed by fire barriers, such as fire resistant walls/floors and fire doors. In the case of a fire incident, the fire barriers, which separate a fire compartment from the adjacent space, will resist the fire spread from the burning fire compartment, thus protecting the rest of the building. A fire compartment may contain several internal spaces such as rooms, hallways and corridors, depending on the compartmentalization of the building.
(2) *Building opening:* Building openings refer to the openings for ventilation or transportation within a building and can be generally classified into three levels, namely the building level (e.g., building exits), the floor level (e.g., floor exits), and the internal level (e.g., room exits). The building opening is usually equipped with a connected component (e.g., doors and windows) or a separated component for fire-fighting purposes (e.g., fire rolling shutters).
(3) *Building components:* Two types of building components, namely the separated component and the connected component, are distinguished according to their effect on space accessibility. Separated components divide the internal space of a building into separated spaces and are able to resist fire spread. It is noteworthy that the dynamic separated components, such as fire doors and fire rolling shutters, are only activated during a fire emergency, while the static separated components such as firewalls and floors always take effect. In contrast to separated components, connected components are building components that connect the adjacent spaces and allow evacuees to transfer from one internal space to another during an evacuation. Typical connected components are doors, windows, staircases and elevators.

4.1.2. Semantics-based topological representation

Based on the pre-defined semantic ontologies in the context of building fire safety, a semantics-based topological representation (see Fig. 2) could be constructed and updated according to the state of the building components. Three basic semantics–based topological relationships are concluded, namely the inclusion relationship, the association relationship and the accessibility relationship.

The inclusion relationship allows the decomposition of buildings into parts according to its semantics-based ontological structure, which follows the structures that are given or can be observed in the real world. For example, a building is generally divided into fire compartments to limit the risk of fire and to prevent the spread of fire and smoke. The internal space of each fire compartment can be further (hierarchically) partitioned into internal spaces such as rooms and corridors, and building components such as doors, firewalls and floors.

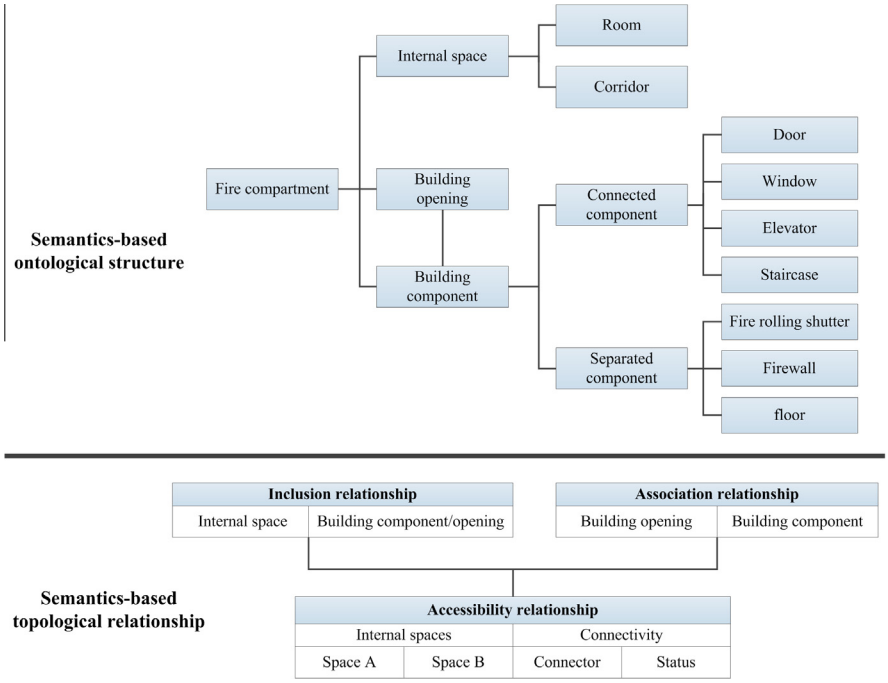


Fig. 2. The semantic ontologies for building fire safety and semantics-based topological relationships.

The association relationship is a semantic relationship that is used to describe the dynamic state of building openings. When a building opening is associated with a connected component such as door, the opening acts as a connected node that allows the transfer from the room to the corridor when the opening is available. However, when a building opening is associated with a dynamic separated component, the opening becomes a dynamic node depending on the state of the separated components. For example, a building opening associated with a fire rolling shutter can be treated as a connected node during normal indoor navigation, but this opening may become an interrupted node in an emergency evacuation as the fire rolling shutter is closed to resist the passage of fire and combustible toxic gasses.

The accessibility relationship refers to the transfer possibilities from an internal space to a neighboring space via a building opening. This relationship determines the availability of the crossing escape route. Neighboring spaces are spaces that include common building openings serving as the connector and, therefore, can be drawn from the inclusion relationship. The accessibility between the neighboring spaces depends on the real connectivity, namely the open/close status of the connector. The accessibility relationship includes vertical accessibility and horizontal accessibility. Vertical accessibility is the accessibility between corridors on different floors via an elevator or a stairway. Horizontal accessibility refers to the accessibility between rooms and corridors on the same floor.

4.1.3. Grid-based representation

A grid-based geometric representation is used to facilitate the simulation of the evacuee's movement within the rooms and corridors. The size of the cells that compose the grid is 40 cm $\times$ 40 cm, which is approximately the typical space that is occupied by a pedestrian in a dense crowd [2]. The cell size may have influence on the simulation results as it determines the maximum density and the flow rates [26,35]. For special situations it might be desirable to use a finer discretization [45]. In this study, 40 cm $\times$ 40 cm cells seem to be sufficient as the model mainly concentrates on knowledge-based route choice. Each cell has the following four attributes: underlying internal space, distance to the openings of the internal space, state of occupancy, and impedance due to the surrounding obstacles.

The agent can be represented using different methods, for instance, the elliptical representation [47] or the three-circle representation [23]. Because the software that we used to implement the model is advanced in spatial analysis but has limitations in representing human body, the agents are assumed to be the same size as the grid cell as in a number of grid-based models.

A cell may be occupied by a stationary obstacle, such as wall and closed door, or by a moving obstacle, namely the agent. Stationary obstacles can be derived from the semantics-based representation and are updated when the dynamic separated building components are activated. Moving obstacles are transferred according to the presence of agents.

People attempt to maintain a distance from obstacles to avoid collisions while moving around. This repulsion force is represented as an impedance value assigned to the neighboring cells of obstacles as shown in Fig. 3 corresponding to formula (1).

$$R_{-1,1} = R_{1,1} = R_{1,-1} = R_{-1,-1} = \frac{\sqrt{2}}{2}F, R_{0,1} = R_{-1,0} = R_{1,0} = R_{0,-1} = F \tag{1}$$

Here, $R_{(i,j)}$ is the impedance value of the cell (i, j) due to the repulsion of the center cell, and F is the average interaction force.

4.2. Knowledge-based evacuation behavior

4.2.1. Spatial knowledge and event knowledge

The evacuee's knowledge of the environment is a decisive factor in determining the escape route. Two types of knowledge, in particular, are significant: spatial knowledge and event knowledge. Spatial knowledge indicates the individual's

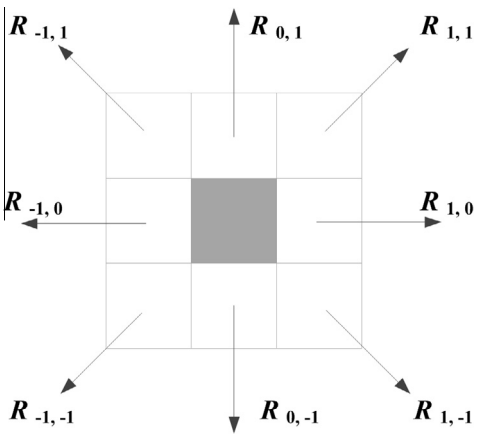


Fig. 3. The repulsion force that is caused by the obstacles in the surrounding cells.

familiarity with the internal structure of the building, namely knowledge about the spatial features of the stationary building environment during a normal situation. Event knowledge refers to the individual's awareness of the dynamic change of the building environment during an emergency situation, and particularly the state of fire safety facilities in this study.

Crucial spatial knowledge for planning an escape route includes the exit location, the spatial accessibility between the internal spaces and the geometric features such as corridor length. The knowledge of the exit location and the spatial accessibility implies a collection of potential escape routes that would be considered by the individual. The knowledge of the geometric features allows the individual to select the most efficient escape route with the shortest traveling distance. Because the individual's familiarity with exits is heavily dependent on the exit utilization, we identified three types of exits: the main exit, which leads to the most possible route that an individual might have followed when entering the building the first time; the normal exit, which is frequently used as an alternate to the main exit during a normal situation; and the emergency exit, which is a protected exit that is enclosed by fire doors and is rarely used during a normal situation.

In a fire emergency, the event knowledge determines whether the evacuee can avoid the blocked routes. Evacuees with event knowledge are aware of the change in the spatial accessibility due to activated fire safety facilities and will accordingly avoid the blocked routes. Otherwise, the evacuee might walk into a blocked route unless he/she encounters the barrier.

Considering the possible person types during a real situation, we classified five representative types of evacuees regarding different levels of prior spatial knowledge and event knowledge, as listed in Table 1.

4.2.2. Individualized escape route planning

During the evacuation, the evacuees perform individualized route planning with different knowledge levels. A diagram of the route planning algorithm is shown in Fig. 4. Initially, the agents are equipped with different levels of prior spatial knowledge and event knowledge. This knowledge could be enriched during the evacuation as the agent may acquire more information of blocked routes, congestion and building exits through observation. During emergency situation, individuals experience psychic stress due to limited time and unexpected danger [37]. A stress threshold was adopted to measure the effect of psychic stress [32]. The agents may proceed with different route selection/adjustment depending on the knowledge level as well as the stress level. The agents with no known exit available will just follow the surrounding agents. The agents with grater psychic stress will choose the most familiar route without considering the travel distance, while the agents with less psychic stress will evaluate all of the familiar escape routes and choose the shortest one.

It is worth noting that evacuees can only see an exit within a certain area [7] as well as an closed fire rolling shutter and a congested area. In the proposed model, a visual observation of the surrounding environment is implemented through a visual sensor based on the ray tracing method [49]. By casting laser rays from the eye position of an agent within a visual angle (e.g., 170°) and visual distance (e.g., 10 m), an agent is able to detect the unknown exits and barriers as well as congestion, and respond to the perceived information (see Fig. 5). For example, as shown in Fig. 5(a), when congestion occurs at the targeted exit, Agent A will redirect to the newly detected exit. In contrast, Agent B will insist on the known exit because the alternative exit is beyond the visual area. In Fig. 5(b), the agent may not notice the front barrier and turn back unless the barrier is within the maximum visual distance.

4.2.3. Local movement

Once the escape route is selected, the agent begins to move toward the selected exit. Local movement within rooms and corridors is implemented through the grid-based representation of the interior spaces. At each time step, the agent moves from a cell to one of the unoccupied neighboring cells. During an unimpeded situation, the agent will move straight forward to the next goal. However, affected by interactions with surrounding obstacles, especially when avoiding collision, the agents are not likely to move strictly along the shortest path. Considering both the agent's intention to move toward the next goal and the influence of the surrounding obstacles, the probability of moving to a neighboring cell (i, j) is calculated by

$$P_{ij} = NI_{ij}(w_d d_{ij} + w_r r_{ij}) n_{ij} \varepsilon_{ij} \tag{2}$$

Here the following factors are considered:

Table 1
Classification of evacuees based on knowledge levels.

Type	Typical individuals	Spatial knowledge			Event knowledge
		Exit	Connectivity	Geometry	
A	Residents with higher fire safety awareness	All	All	All	Yes
B	Residents with lower fire safety awareness	All	All	All	No
C	Regular visitors	Main exit, normal exit	Partial	Partial	No
D	First-time visitors who know the route they came	Main exit	Partial	Partial	No
E	First-time visitors who forget the route they entered	None	None	None	No

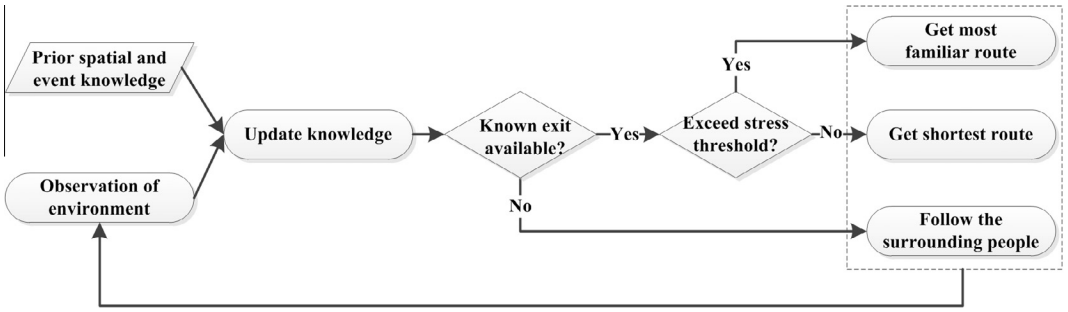


Fig. 4. The escape route planning diagram.

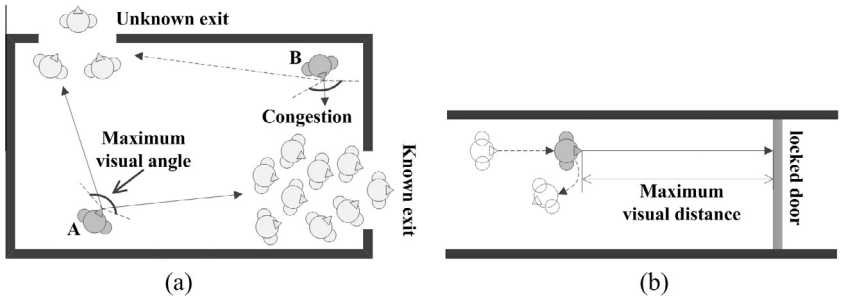


Fig. 5. The agent's visual observation of the surrounding environment.

(1) distance to the goal:

$w_d d_{ij} = w_d(\max_{D_{ij}} - D_{ij})/(\max_{D_{ij}} - \min_{D_{ij}})$, where D_{ij} is the distance from cell (i, j) to the agent's goal, and $w_d > 0$ is the weight value of the distance factor.

(2) repulsion from obstacles:

$w_r r_{ij} = w_r(\max_{R_{ij}} - R_{ij})/(\max_{R_{ij}} - \min_{R_{ij}})$, where R_{ij} is the repulsion force in cell (i, j) that is received from the surrounding obstacles, and $w_r > 0$ is the weight value of the repulsion factor.

(3) occupancy by dynamic obstacle (agent):

$$n_{ij} = \begin{cases} 0 & \text{for cells occupied by other agents} \\ 1 & \text{else} \end{cases}$$

(4) occupancy by stationary obstacle (building obstruction):

$$\varepsilon_{ij} = \begin{cases} 0 & \text{for occupied cells, e.g. walls} \\ 1 & \text{else} \end{cases}$$

(5) inertial enhancement:

$$I_{ij} \begin{cases} > 1 & \text{for cells in the previous movement direction} \\ = 1 & \text{else} \end{cases}$$

(6) normalization:

$$N = \left[\sum I_{ij}(w_d d_{ij} + w_r r_{ij}) n_{ij} \varepsilon_{ij}\right]^{-1}$$

5. Agent-based simulation of building evacuation and the evacuation efficiency analysis

Using the evacuation model that was presented in the previous section, we conducted evacuation simulations of the second floor of a campus building as a case study. However, because the evacuees' movement in a staircase is quite complex [38] and is beyond the focus of this study, we simulated only the evacuee's movement toward the staircase and assumed that the agent will leave the floor once it reaches an open staircase or reaches the door connecting to a staircase. Accordingly, to simplify the situation and concentrate on the influence of people's awareness of the predictable spatial change, we assumed that all of the agents can select the shortest route from the collection of known routes. We used Agent Analyst [17], which is an open-source software compatible with the widely used ArcGIS software, as the simulation platform.

5.1. Environment and settings

The floor plan includes seven lecture rooms, six staircases, three elevators and two doors leading to an outdoor staircase, as shown in Fig. 6. Three fire rolling shutters are installed in the corridor and divide the floor plan into four fire compartments when shutting off. Considering the fact that the evacuees are not allowed to use elevators during a fire emergency, the three elevators are set as occupied space. The equipment rooms and storage rooms are also considered occupied space because these rooms are usually locked, and people rarely go inside.

There are eight exits on the floor plan. Exit 2 is an open staircase and is adjacent to the most frequently used elevator. Therefore, we assume Exit 2 as the main exit of the floor. Exit 4 and Exit 6 are doors connecting to an outdoor staircase and Exit 8 is an open staircase that is located in a hallway. These three exits are normal exits that are frequently used by the students who are familiar with the building. Exit 1, Exit 3, Exit 5, and Exit 7 are emergency exits that are enclosed by fire doors connecting to the staircases and are rarely used during a normal situation. Accordingly, the known exits and situational awareness for each type of evacuee that is distinguished in Section 4 are listed in Table 2.

Because this study mainly concentrates on the influence of the predictable spatial accessibility that is changed by activated fire safety facilities, the situation is to some extent simplified. For available exits, the doorway-leaves are assumed to be fully extended. The effective doorway width and its influence on crowd flow that have been discussed in previous studies [13,14] are not presented here. In addition, the impact of environmental conditions (e.g., smoke and toxic gasses) on individual's route choice, which is mainly related to the evacuee's physical fitness and psychological characteristics, is not included in this case study. With these assumptions, three possible fire scenarios were designed when different fire rolling shutters were activated. The fire scenarios are as follows:

- *Scenario One:* A minor fire incident occurs on the floor and no fire rolling shutter is shut off.
- *Scenario Two:* A major fire incident occurs within the second fire compartment, and the fire rolling shutters 201 and 202 are shut off to prevent the fire from spreading.
- *Scenario Three:* A major fire incident occurs within the third fire compartment, and the fire rolling shutters 202 and 203 are shut off to resist fire spreading.

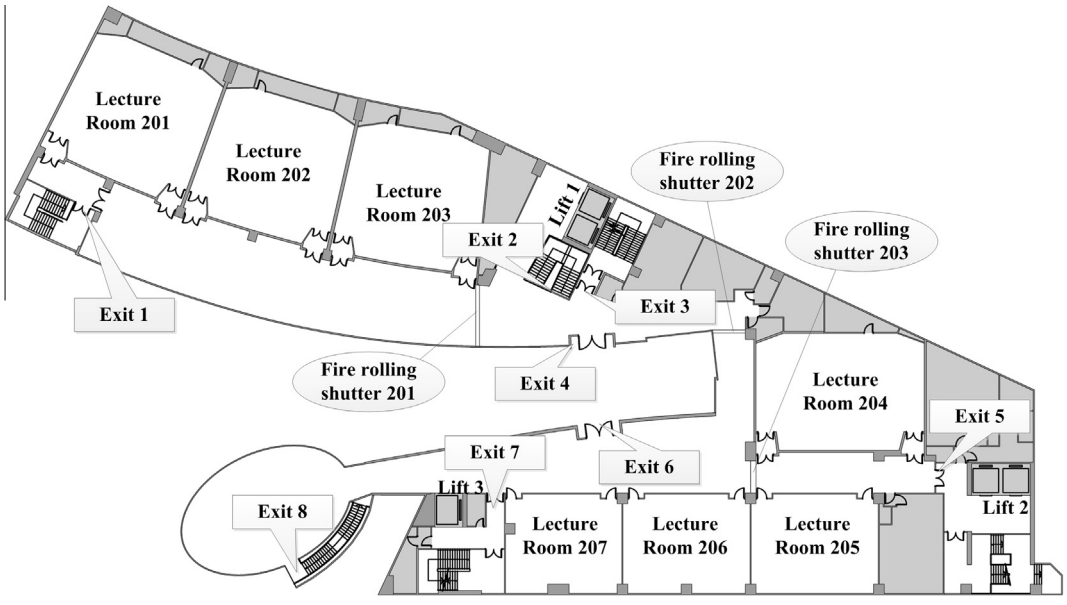


Fig. 6. The floor plan of the second floor of a campus building.

Table 2
The knowledge levels of the different types of agents.

Agent type	Known exits								Fire rolling shutter
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	
A: students with higher fire safety awareness	√	√	√	√	√	√	√	√	√
B: students with lower fire safety awareness	√	√	√	√	√	√	√	√	
C: students without fire safety awareness		√		√		√		√	
D: visitors knowing the main exit		√							
E: visitors knowing none of the exits									

Because the construction of the evacuees greatly influences the evacuation efficiency [21], we also designed three groups of evacuees for different situations (see Table 3). The first group is designed to simulate an optimal situation when all of the evacuees are aware of the possible escape routes and the state of the fire rolling shutters. The second group is designed to simulate an emergency evacuation during a normal day, when students represent the majority of the evacuees. The third group is designed to simulate a particular situation when a conference is being held on this floor and when most evacuees are first-time visitors. It is worth noting that the simulated evacuees will start to escape at the initial time. Pre-movement time, which may vary due to different types of alarm during a class room evacuation [39], are not included in this experimental case study.

5.2. Potential escape routes

The semantic representation of the floor plan is constructed and contains the inclusion relationship, the connectivity relationship, and the accessibility relationship. Using Lecture Room 206 as an example (see Tables 4–6), this room includes the separated components Firewall 215, Firewall 216, Firewall 217 and Firewall 218; the connected components Door 211 and Door 212; and the building openings Opening 225 and Opening 226. Opening 225 and Opening 226 are associated with Door 211 and Door 212, enabling the accessibility relationship between Lecture Room 206 and Corridor 205 as they are opening. Similarly, the accessibility relationship between Corridor 205 and Corridor 204 depends on the state of Fire Rolling Shutter 203. When Fire Rolling Shutter 203 is closed, the connectivity relationship between Corridor 204 and Corridor 205 becomes invalidated, making Exit 5 no longer accessible to Lecture Room 206.

The potential escape routes for each type of agent are derived from the semantic representation in accordance with the agent's knowledge level. Considering the eight escape routes from Lecture Room 206 as an example (see Fig. 7), the potential routes from Lecture Room 206 that will be considered by each type of agent at the initial time, including both blocked and

Table 3
The construction of the different groups of evacuees.

	Type A		Type B		Type C		Type D		Type E		Total
	%	No.	%	No.	%	No.	%	No.	%	No.	
Group 1: optimal situation	100	680	0	0	0	0	0	0	0	0	680
Group 2: normal day	10	68	30	204	40	272	10	68	10	68	680
Group 3: conference day	5	34	5	34	10	68	40	272	40	272	680

Table 4
The instances of the inclusion relationship.

Internal space	Building component/opening
Lecture Room 206	Firewall 215. . ., Door 211. . ., Opening 225. . ..
Corridor 204	Firewall 212. . ., Fire Door 205. . ., Opening 230. . ., Shutter 203
Corridor 205	Firewall 215. . ., Door 211. . ., Opening 230. . ., Shutter 203

Table 5
The instances of the association relationship.

Building opening	Building component
Opening 225	Door 211
Opening 226	Door 212
Opening 230	Shutter 203
Opening 217	Fire Door 205

Table 6
The instances of the accessibility relationship.

Internal space		Connectivity	
Space 1	Space 2	Connector	Status
Lecture Room 206	Corridor 205	Opening 225	Open
Lecture Room 206	Corridor 205	Opening 226	Open
Corridor 204	Corridor 205	Opening 230	Open/close

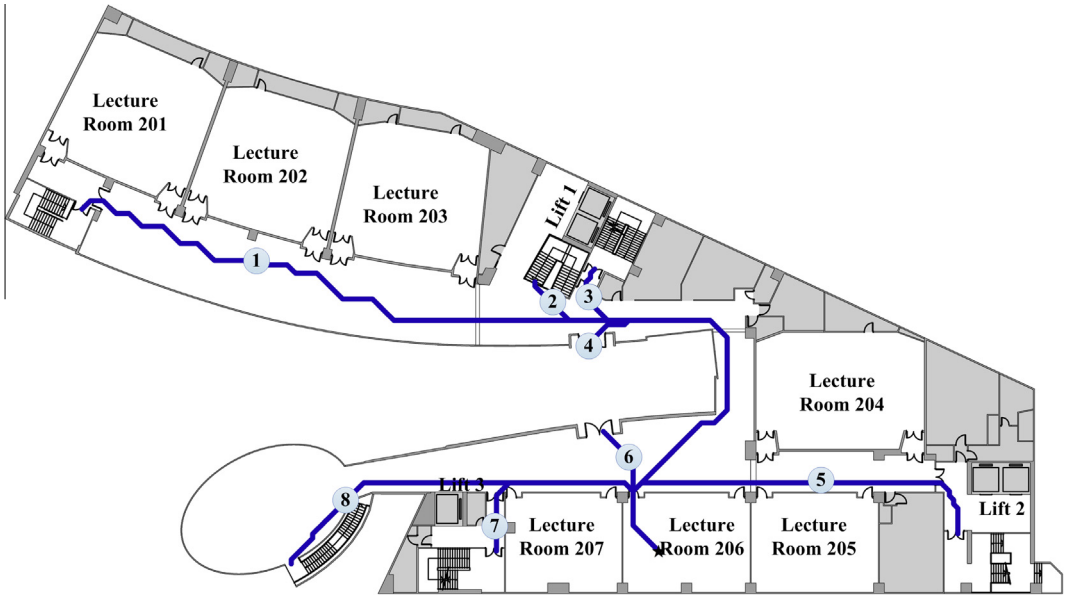


Fig. 7. The potential escape routes from Lecture Room 206.

Table 7
The potential escape routes for each type of agent at the initial time.

Agent type	Scenario One	Scenario Two		Scenario Three	
		Unblocked	Blocked	Unblocked	Blocked
A	1, 2, 3, 4, 5, 6, 7, 8	5, 6, 7, 8	None	6, 7, 8	None
B	1, 2, 3, 4, 5, 6, 7, 8	5, 6, 7, 8	1, 2, 3, 4	6, 7, 8	1, 2, 3, 4, 5
C	2, 4, 6, 8	6, 8	2, 4	6, 8	2, 4
D	2	None	2	None	2
E	None	None	None	None	None

unblocked routes in Scenario Two and Scenario Three, are listed in Table 7. During the evacuation, the agent will eliminate the blocked routes and turn back as it encounters the closed fire rolling shutter, and new routes will be added to the agent's route options when the initially unknown exits are observed.

5.3. Results and analysis

To compare the attractiveness of the exits and the evacuation times of different groups of agents during the three scenarios, a counter was planted at each exit to count the agents passing through. The overall utilization of the exits is shown in Fig. 8, and the collected data are illustrated as a function of the elapsed time in Figs. 9-11. From the simulation results, the following observations are noticed.

The first observation is that the utilization of the emergency exit highly depends on its visibility. The agents without a complete knowledge of the exit location tend to ignore the emergency exit outside of their visual area. This occurrence is particularly true for Exit 7, which is parallel to the central axis of the corridor and on the same side of the room doors. During all of the scenarios, the utilization of Exit 7 by Group 2 and Group 3 is very low (see Fig. 8) because Exit 7 is hardly noticed by the agents who are not familiar with the exit. In contract, Exit 1, Exit 3 and Exit 5 attract more agents due to better visibility.

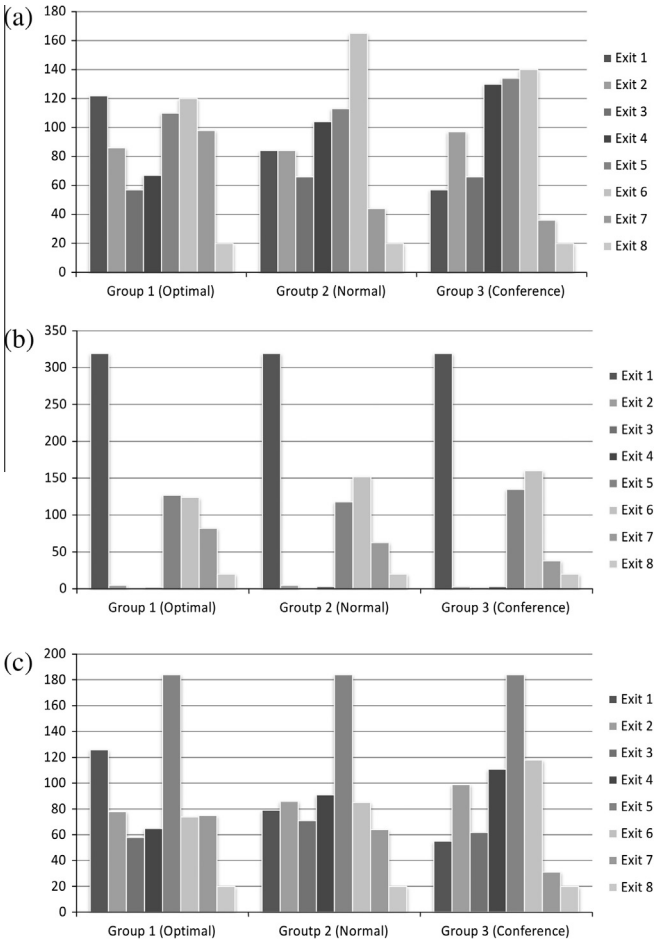


Fig. 8. The utilization of each exit with different groups of evacuees during Scenario One (a), Scenario Two (b), and Scenario Three (c).

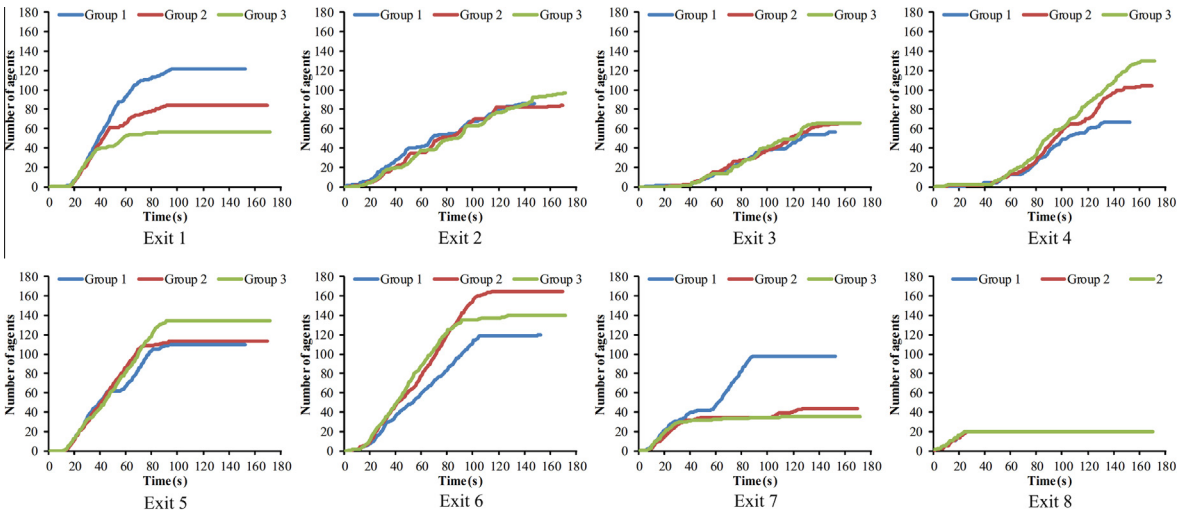


Fig. 9. The cumulative agents passing through each exit during Scenario One for each group of evacuees.

Second, during Scenario Two, the evacuation times of Group 2 and Group 3 are significantly longer than the evacuation time of Group 1 (see Fig. 10). This occurrence is mainly due to a lack of event knowledge. In Group 2, a number of agents in Lecture Room 201, Lecture Room 202 and Lecture Room 203 are not aware of the fact that Fire Rolling Shutter 201 is shut off

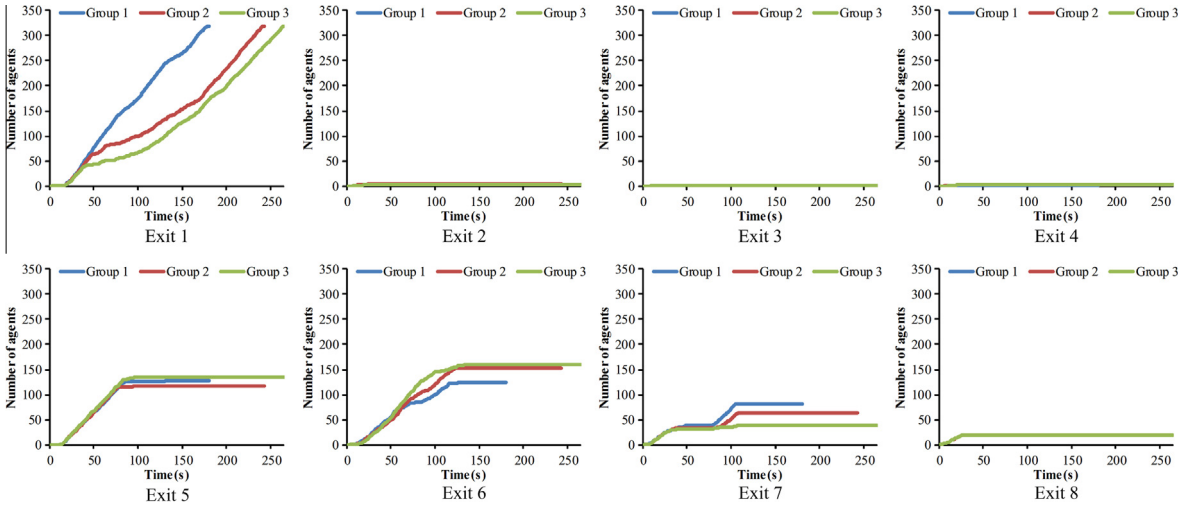


Fig. 10. The cumulative agents passing through each exit during Scenario Two for each group of evacuees.

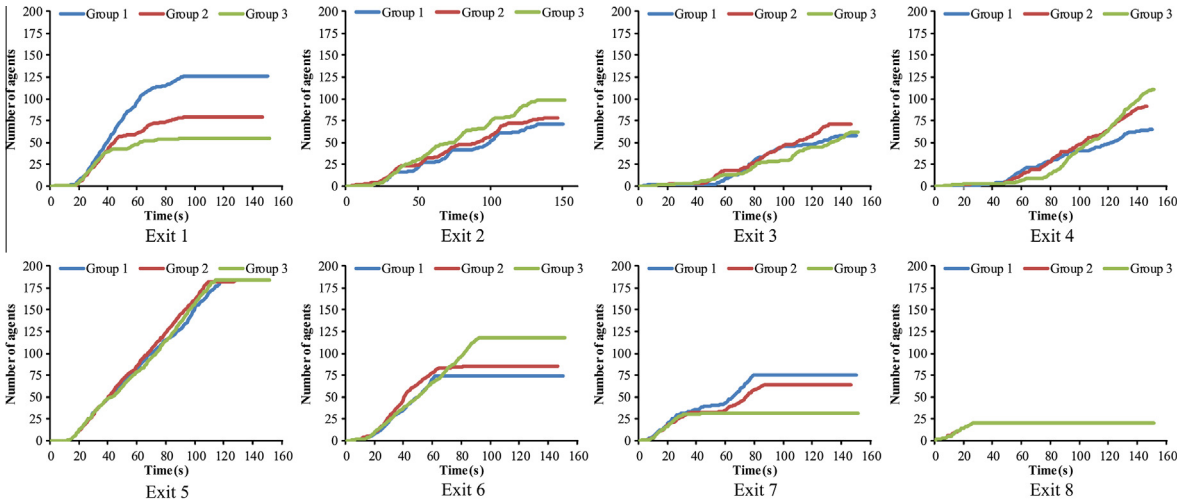


Fig. 11. The cumulative agents passing through each exit during Scenario Three for each group of evacuees.

and that the routes toward Exit 2, Exit 3 and Exit 4 are consequently blocked. These agents continue to move toward the unavailable exits until they perceive the closed door and turn back to Exit 1. When congestion occurs at Exit 6, the agents may redirect to Exit 4 if they consider that Fire Rolling Shutter 202 is still opening. Such detouring behavior results in the lag of outflow through Exit 1 and Exit 6 compared with that of Group 1. The problem is worse for Group 3, as more agents do not initially have event knowledge.

However, the lack of event knowledge does not always lead to detouring behavior and does not always influence the evacuation efficiency when the spatial connectivity is changed. In Scenario Three, the evacuation times of Group 1, Group 2 and Group 3 are quite similar even though the spatial connectivity was changed by Fire Rolling Shutter 202 and Fire Rolling Shutter 203 (see Fig. 11). The agents in Lecture Room 204 and Lecture Room 205 are forced to use Exit 5 without any chance for detouring. The agents in Lecture Room 206 and Lecture Room 207 evacuate from Exit 6 or Exit 7 without making a detour to Exit 4, because Exit 6 and Exit 7 are apparently the best options and congestion is not likely to happen with such an amount of outflow. Therefore, the impact of the spatial change on the evacuation time is not only related to the evacuees' knowledge level but also to the location of the fire rolling shutter.

Last but not least, unreasonable route re-planning increases the evacuation time, but is sometimes prevented when the fire rolling shutter is activated. A typical example of unreasonable route re-planning is redirecting from Exit 1 to Exit 2, Exit 3 or Exit 4. For all of the groups of agents during Scenario One and Scenario Three, the agents from Lecture Room 202 and Lecture Room 203 tend to opt to Exit 2, Exit 3 or Exit 4 when congestion occurs at Exit 1. However, this re-planning causes an even longer evacuation time as the evacuation at Exit 1 stops much earlier than Exit 2, Exit 3 and Exit 4 (see Fig. 9).

Furthermore, during the evacuation of Group 2 and Group 3 during Scenario One, when Exit 6 is congested by the agents from Lecture Room 204, Lecture Room 205, Lecture Room 206 and Lecture Room 207 due to an unawareness of Exit 5 and Exit 7, those agents may turn to the frequently used Exit 2 or Exit 4, resulting in an even longer evacuation time than waiting to use Exit 6. However, the unreasonable behavior of redirecting from Exit 6 to Exit 2 or Exit 4 is eliminated in Scenario Three by closed Fire Rolling Shutter 203 as previously discussed. Therefore, the spatial change during Scenario Three positively influences the evacuation efficiency by balancing the utilization between Exit 5 and Exit 6.

6. Conclusions and implications

This paper presents an agent-based building evacuation model that is able to simulate individualized evacuation behaviors resulting from the differing knowledge levels of the building's internal structure and simulate the predictable spatial accessibility that is changed by the activated fire safety facilities. In the proposed model, the building environment is represented by a semantic model at the macro level, where the connectivity relationship between the internal spaces, such as rooms and corridors, is updated according to the state of the fire safety facilities. At the micro level, a grid-based representation is adopted to facilitate the simulation of the evacuees' movement within the internal space. The evacuees are modeled as autonomous agents that are equipped with not only the spatial knowledge of the stationary spatial features but also the event knowledge of the activated fire safety facilities. Therefore, the model makes it possible to simulate a more realistic movement of the evacuees during different fire scenarios when the specific fire safety facilities are activated, considering the evacuee's awareness of the corresponding change in the spatial accessibility.

Using the proposed model, we conducted a series of evacuation simulations of the second floor of a campus building during three specific fire scenarios with different groups of evacuees. The simulation results indicate that the predictable spatial change might positively affect the improving evacuation efficiency. It is also crucial to identify the potential negative influence of the fire safety facilities and to provide necessary guidance to prevent detouring during an emergency evacuation. The proposed model demonstrates an approach to evaluate the potential influence of the activated fire safety facilities on the evacuation efficiency, thereby providing a more rational basis for evacuation management and useful suggestions for designing the building structure and installing the fire safety facilities.

However, it is important to note that the proposed model mainly concentrates on the influence of the predictable change in the spatial accessibility and does not include all of the factors that have or might have an effect on the spatial accessibility of the building environment. As suggested by a number of studies [10,12], fire smoke and toxic gasses will result in dynamic spatial accessibility and impact evacuees' moving speed and direction. While the influence of activated fire safety facilities is related to the evacuees' spatial and event knowledge, the influence of smoke and toxic gasses is more related to the evacuees' physical fitness and psychological characteristics. Therefore, in the future work we will consider the evacuees' reaction when encountering the smoke in addition to their capability of avoiding the route that is blocked by activated fire safety facilities. Another issue is about the cell size of grid-based representation. In this study, we used 40 cm $\times$ 40 cm cells given the typical area occupied by a pedestrian and the computational efficiency. As we have mentioned, the cell size may influence the simulation results. The sensitivity of simulation results to the cell dimension needs to be investigated in the future. Besides, when modeling the evacuees' movement at the micro level, we have made simplifications and some factors were not fully considered, such as the representation of human body and the effect of dynamic effective width of exit. Given these facts, the model presented here is a prototype. More efforts are in need to validate the model and make it more applicable in the future.

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